

Bug Fix Log – BITS Digital CodeForge V2.0

#	Bug / Issue Identified	How it was Reproduced	Root Cause	Fix Implemented	How the Fix was Tested
1	Course list duplicates after uploading another file	Uploaded an initial Excel file, then sequentially uploaded a second Excel file without refreshing.	The application appended new course options to the dropdown without clearing the existing options array first.	Added a reset mechanism to clear existing course options before populating options from the newly uploaded workbook.	Uploaded two different Excel files sequentially and verified that the course dropdown cleanly displays only the unique courses corresponding to the active file.
2	Average shows an incorrect value when no students are present	Selected a course/program category that contained zero student entries or empty records.	The average calculation logic executed directly on an empty dataset, dividing the cumulative score sum by zero, resulting in NaN.	Implemented defensive checking to evaluate array lengths, returning safety fallbacks and gracefully handling empty datasets before calculations run.	Selected a course with no valid records and confirmed that statistics safely display a dash instead of breaking or showing NaN.
3	Timer clock visually delays on initialization	Loaded the application interface and observed the timer display element.	The countdown function only initialized updates via setInterval, forcing the clock to sit blank at 00:00 until the first 1-second interval tick fired.	Added an immediate initial execution call right inside DOMContentLoaded before setting up the recurring interval.	Refreshed the page and verified that the timer immediately renders 00:00 on screen without any noticeable initial lag.

Summary of Enhancements (Stage 2 & 3)

- **Robust Data Sanitization:** Added safety handling for messy imports, decimal rounding, and out-of-bounds score clamping.
- **Live Analytics & Visualization:** Enhanced the histogram feature with smooth curve fits and interactive click-to-filter grade roster states.
- **Cloud & Deployment Ready:** Integrated Firebase configuration framework setup and responsive multi-attempt submission timers.